

Structural Hallucination in Large Language Models: A Network-Based Evaluation of Knowledge Organization and Citation Integrity

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Abstract

Large Language Models (LLMs) increasingly mediate access to scholarly information, yet their outputs are typically evaluated at the level of individual statements rather than knowledge structure. This paper introduces *structural hallucination*: systematic distortion of conceptual organization, relational architecture, and bibliographic grounding that remains invisible to sentence-level accuracy metrics. To detect such distortions, we develop a network-based hallucination stress test grounded in knowledge graph extraction, graph similarity analysis, centrality comparison, and citation integrity verification.

The protocol is applied to three structured domains representing core forms of scholarly knowledge: Roget’s *Thesaurus* (1911) as a historical knowledge organization system; Wikidata philosophers as a biographical knowledge graph; and bibliographic citation records retrieved from the Dimensions.ai database. Across all domains, substantial structural divergence is observed. In the lexical benchmark, macro-averaged F1 scores fall below 0.05; in the biographical benchmark, hallucination rates exceed 93%; and in the bibliometric benchmark, citation omission reaches 91.9%. Network-level comparison in the Roget reconstruction further reveals node-set Jaccard similarity of 0.028 and source-mismatch rates above 94%.

These findings show that structural fidelity cannot be inferred from local fluency alone. The proposed stress test provides a reproducible instrument for evaluating the structural integrity of LLM-generated knowledge representations within knowledge organization and information quality research.

Keywords: large language models; structural hallucination; knowledge organization; knowledge graphs; network analysis; information quality; bibliometrics; citation analysis

1 Introduction

Large language models (LLMs) increasingly mediate access to scholarly information, but fluent output need not preserve the concepts, relations, or bibliographic links that organize a domain. This study examines that gap through *structural hallucination*: relational mismatch that is not adequately captured by sentence-level accuracy measures. We develop a network-based stress test that represents selected reference sources and LLM outputs as comparable graphs, aligns them under benchmark-specific rules, and assesses node–edge overlap, network diagnostics, and citation integrity.

The framework is applied to the 1911 edition of Roget’s *Thesaurus*, Wikidata philosopher records, and Dimensions citation records. It asks whether the specified model and procedures recover the selected source structures, not whether all LLMs or knowledge organization systems behave alike. The contributions are a source-bounded account of structural hallucination, a reproducible comparison protocol, and evidence from three distinct benchmarks.

2 The Epistemic Problem of Fluency, Error, and Accuracy

LLMs produce plausible continuations rather than verified retrievals (Bender et al., 2021; Bommasani et al., 2021). Their output can therefore be grammatical and contextually coherent while containing factual, bibliographic, or relational error. Hallucination taxonomies commonly distinguish intrinsic and extrinsic proposition-level falsehood (Huang et al., 2025; Ji et al., 2023; Tonmoy et al., 2024); the present study addresses the additional question of whether generated material preserves an externally documented relational structure.

This distinction complements recent evaluation of LLM-generated literature reviews. Tang, Duan, and Cai (Tang et al., 2025) evaluate reference generation, literature-summary writing, and review composition, including generated-reference hallucination, semantic coverage, and factual consistency against human-written counterparts. The present framework instead evaluates alignment between generated structured representations and designated reference sources, using graph and network diagnostics. Fact-checking remains necessary, but verifying isolated claims cannot establish whether a resulting representation retains the conceptual or bibliographic organization of a field.

3 Structural Hallucination: Definition and Mechanisms

3.1 Defining Structural Hallucination

Structural hallucination is a mismatch between the relational architecture represented in generated material and that represented in a documented comparison source. A generated statement may be locally plausible or correct while the combination of entities, conceptual dependencies, intellectual genealogies, citation links, or relation directions fails to recover the selected source structure. The term therefore concerns configuration, not merely proposition-level falsehood (Ji et al., 2023; Maynez et al., 2020; Rawte et al., 2023).

Three related manifestations motivate the analysis: conceptual re-centering, in which peripheral concepts are assigned undue structural prominence; bibliographic distortion, including source-nonmatched, omitted, or misattributed citations; and logical mis-structuring, in which locally plausible claims imply unsupported relations. These mechanisms can coexist and should be evaluated against an explicit source, graph-construction procedure, and matching rule. The following sections specify that evaluation design.

4 Governance as Methodological Infrastructure

4.1 From Governance to Reproducible Validation

Here, governance is used in a restricted methodological sense: a validation design that makes generated representations inspectable against documented reference structures. It combines graph extraction and direct node–edge comparison, network diagnostics, and citation verification. Direct comparison records shared, missing, and generated-only entities or relations; network measures then describe differences in centrality and community structure. Citation checks examine reference existence, metadata, and omission relative to the selected record. These components are complementary: they do not convert a reference source into absolute truth, but they make the scope and basis of comparison explicit.

The subsequent benchmarks apply this design to lexical, biographical, and bibliographic sources. Their purpose is to test structural alignment under stated prompts, extraction procedures, matching rules, and operational reference sources.

5 Knowledge Graphs and Network-Analytic Governance

5.1 Reference Structures, Extraction, and Comparison

A knowledge organization system (KOS) is a curated conceptual and organizational scheme whose scope, provenance, representational rules, and community use shape a domain; a knowledge graph (KG) is a graph-based representation of entities or concepts and typed relations (Hjørland, 2003, 2021; Zeng, 2008). The two categories are distinct but overlapping. A KOS may be implemented or exported as a KG, and a KG may incorporate a KOS, but neither term is automatically synonymous with the other. Here, the reference graph is a documented export of, or a transparent derivation from, the selected source; the LLM graph is an analytic representation extracted from generated text. The comparison therefore evaluates two documented graph representations, not interchangeable categories or an absolute truth claim.

The reference source determines what the analysis can establish. Its scope, version, curatorial choices, omissions, and biases must be reported. When no canonical graph exists, researchers may use a transparent expert reference, multiple documented references, or a consensus procedure and report how conclusions depend on that choice. Information quality (IQ) is considered here at the relational level: accuracy, completeness, and consistency of graph structure rather than isolated attributes (Arazy and Kopak, 2011; Stvilia et al., 2007).

5.2 Matching and Network Diagnostics

The analysis first extracts entities and typed relations from the generated material and the selected reference source, then aligns records under a benchmark-specific key. Node equivalence is established before graph intersection, difference, or centrality comparison. For generated node u and reference node v , let A_b indicate aligned records, ρ indicate compatible fields or relation roles, and E_b indicate the benchmark-specific identifier-or-lexical test. The matching rule is

$$M_b(u, v) = 1 \iff A_b(r(u), r(v)) = 1 \wedge \rho(u) = \rho(v) \wedge E_b(u, v) = 1.$$

Roget uses exact normalized strings; Wikidata and Dimensions use the thresholds specified in the empirical pipeline. This strict operational matching is deliberate: it makes the comparison reproducible and auditable, prevents an unreported similarity model from silently converting non-identical values into matches, and tests whether the generated output recovers the designated reference structure under a stated schema. Candidate-generation aids such as semantic similarity may support review but do not by themselves establish a match.

The resulting node–edge comparison identifies shared, missing, and generated-only elements. Network diagnostics then compare centrality, community structure, and graph similarity among the documented graphs. This order separates what is present or absent under the matching rule from how retained and generated structures are organized.

5.3 Running Example: Kuno Fischer

For Kuno Fischer, the cached Wikidata reference record gives birth date 1824-07-23; the stored reconstruction gives 1824-04-22. Under the local date rule, the values share only the year token and do not match. This illustrates a source-derived and a generated attribute edge that diverge before any graph-level measure is calculated; it does not itself classify the generated value as a hallucination.

The implementation checklist, full notation key, and pseudocode are supplied in Supplementary Appendix S1.

6 Citation Integrity and Bibliometric Grounding

Citation integrity is the bibliographic component of structural evaluation. Generated references should be checked against authoritative bibliographic sources for existence, metadata, and the relevance of the cited work to the claim it supports ([Alkaissi and McFarlane, 2023](#); [Orduña-Malea and Cabezas-Clavijo, 2023](#)). At the structural level, the generated citation graph can be compared with a documented reference graph for missing works, altered citation neighborhoods, centrality differences, and uneven author or venue representation. These analyses assess relational alignment rather than local correctness alone.

7 Applications: Assessment, Manuscript Writing, and Syllabus Design

The same comparison principle can support source-aware assessment, literature-review checking, and AI-assisted metadata or syllabus design. In each case, the appropriate comparison source, scope, version, and matching rules must be explicit. The framework is intended as a verification method, not as a substitute for domain judgment or a claim that one reference source exhausts valid knowledge.

8 Empirical Benchmarks: Three Stress Tests of Structural Hallucination

Shared Evaluation Pipeline

The three benchmarks use source-specific operational reference baselines, not exhaustive accounts of all valid knowledge. For each selected reference record, `gpt-4.1-mini` was prompted to reconstruct the relevant structured fields in JSON. Responses were parsed into the reference schema, cached after the first successful call, and retried with exponential backoff when necessary. The evaluated model is the research object; the manuscript’s revision-assistance disclosure appears in the end matter.

Records were aligned by normalized Head name (Roget), philosopher name (Wikidata), or publication title (Dimensions). Alignment does not establish value equivalence. Values were compared only within the aligned record, field, and relation role. The implementation does not perform one-to-one assignment for list-valued fields: a value is classified by whether at least one eligible match exists. Multiple eligible values require a documented tie-break or review before one-to-one structural comparison.

Roget uses lower-cased, stripped exact membership. Wikidata uses lower-cased, stripped values and accepts containment or shared-token proportion at least 0.60; a local nonmatch is checked against DBpedia, yielding either verified extra knowledge or a value absent from both sources. Dimensions uses lower-cased, trimmed token-set Jaccard similarity of at least 0.60, while DOI resolution, citation-title similarity, year match, and DOI validity remain separate diagnostics. Precision, recall, F1, and accuracy follow these benchmark-specific procedures. The available artifacts contain no threshold sensitivity analysis or alternative assignment rerun, so all results are conditional on the stated matching rules.

Semantic similarity uses `all-MiniLM-L6-v2` as complementary contextual evidence, not as an equivalence decision. The exploratory logistic-regression classifier is similarly descriptive of operational labels rather than an independent reference baseline.

8.1 Benchmark 1: Lexical Reconstruction of Roget’s Thesaurus

The first benchmark uses the 1911 edition of Roget’s *Thesaurus of English Words and Phrases* (Roget, 1911), obtained from Project Gutenberg. Roget’s *Thesaurus* is a historically situated, hierarchically organized lexical knowledge organization system (thesaurus) that arranges English

words and phrases into Classes, Sections, and Heads. Each Head functions as a lexical-semantic grouping and is associated with curated lists of nouns, verbs, adjectives, adverbs, and cross-references to related Heads. The 1911 edition contains 1,000 Heads.

Source extraction, sampling, and generation. We treated the 1911 edition as a historical reference source rather than an unqualified reference baseline. A custom Python parser processed Project Gutenberg transcription No. 10681, identified numbered Head boundaries, excluded sub-entries and malformed or empty records, and returned 997 usable Heads. It extracted noun, verb, adjective, and adverb sections as semicolon-split, trimmed list strings; cross-references came from `&c <number>` markers and were mapped to Head names where possible.

With Python seed 42, we sampled 30 Heads without replacement. The routine obtained one `gpt-4.1-mini` JSON response per Head (or reused a cached response), with temperature 0 and a 1,024-token maximum. The archived run has 30 successful records but no request timestamp or full provider metadata. The prompt requested `class`, `section`, `head_name`, the four part-of-speech lists, and `cross_references`; the fixed prompt is reproduced in Supplementary Appendix S1.

Reference and generated records were aligned by normalized `head_name`. For each aligned field, lower-cased, stripped list strings were compared by exact set membership; missing or non-list values were empty sets. Precision, recall, and F1 pool TP, FP, and FN across the 30 Heads, so they are micro exact-string scores by field. Synonymy, inflection, aliases, and semantic similarity did not establish matches; generated-only and reference-only strings are therefore protocol-specific mismatch categories, not factual-falsehood verdicts.

8.1.1 Classical Evaluation

Classical evaluation, using exact membership in normalised term sets, indicates extremely low reconstruction fidelity ($F1 < 0.05$) across all fields (Table 1).

Table 1: Classical evaluation results for Roget’s Thesaurus reconstruction ($n = 30$ Heads). All scores are pooled across Heads.

Field	Precision	Recall	F1	Accuracy
noun_list	0.024	0.020	0.022	0.00
verb_list	0.019	0.020	0.020	0.00
adjective_list	0.017	0.019	0.018	0.00
adverb_list	0.000	0.000	0.000	0.00
cross_references	0.038	0.053	0.044	0.00

LLM vs Roget's Thesaurus (n=30)

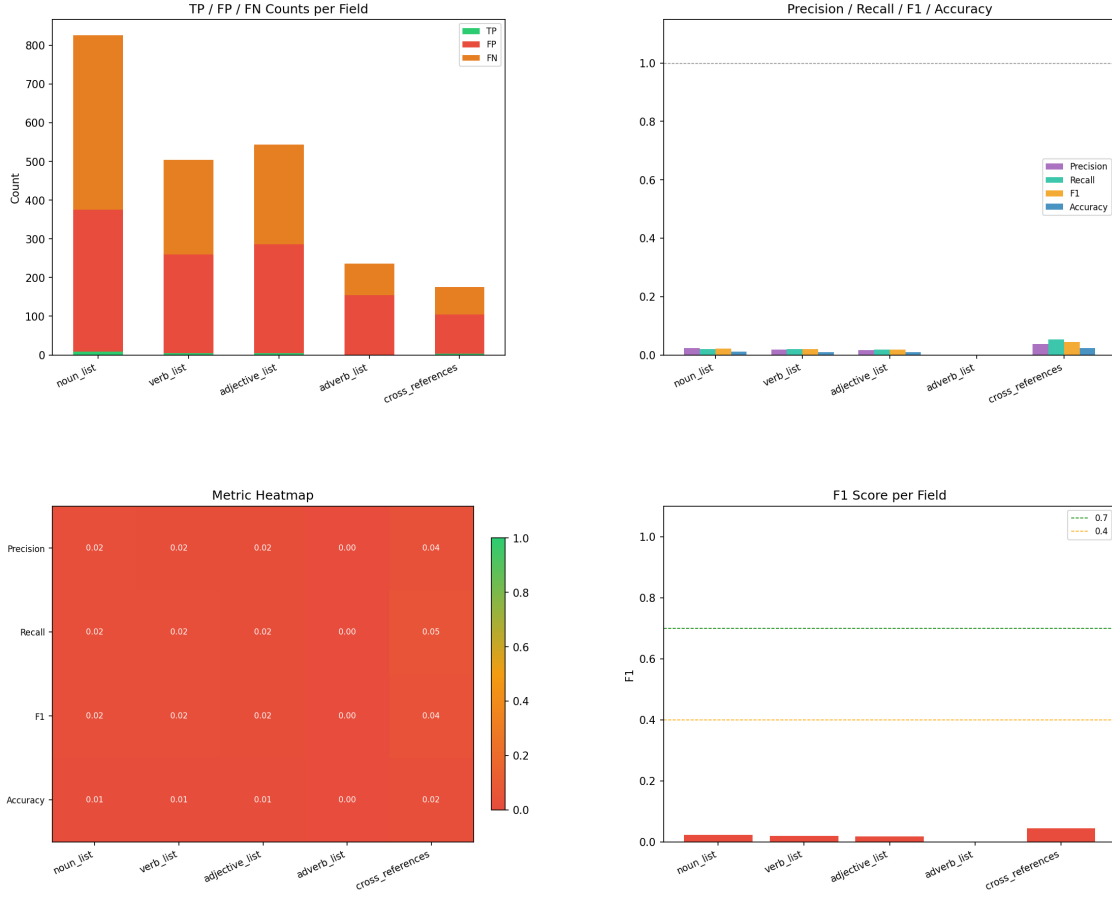


Figure 1: Classical evaluation of LLM reconstruction of Roget’s Thesaurus ($n = 30$ Heads). The four panels show (top left) TP/FP/FN counts per field, (top right) precision, recall, F1, and accuracy per field, (bottom left) metric heatmap across all fields, and (bottom right) F1 score per field with reference thresholds at 0.4 and 0.7. The dominant category in the TP/FP/FN panel is false negatives, reflecting the model’s near-total inability to recall the 1911 vocabulary. All F1 scores are below 0.05; adverbs score exactly zero.

Across the five fields, pooled exact-string F1 ranges from 0.000 (adverbs) to 0.044 (cross-references). The result is a low recovery of the terms and links recorded in the 1911 edition under the specified exact matching protocol. It does not determine whether every generated-only term is semantically unrelated or factually false, and historical lexical change is a possible contributor to mismatch.

Representative cross-reference cases. For *Greatness* (Head 31), the reference lists *Infinity*, *Importance*, *Superiority*, *Multitude*, *Completeness*, and *Wonder*; the cached reconstruction returns

Power, *Influence*, *Eminence*, and *Excellence*, yielding zero exact matches. For *Permission* (Head 760), the reference relation is *Consent*; the model returns *Law*, *Authority*, *Power*, and *Consent*, yielding one exact match and three model-only values. These are mismatches from the structure recorded in the 1911 edition, not proof that every generated relation is false.

Supplementary Figures S1–S2 report the Roget list-level semantic-similarity and exploratory classifier analyses. They are complementary descriptive evidence and were not used to establish exact matches or change the reported exact-string counts.

8.1.2 Network-Analytic Hallucination Stress Test

Graph Structure and Source-Mismatch Rate. The directed knowledge graphs constructed from the Roget and LLM data exhibit dramatically different structural properties (Table 2).

Table 2: Structural comparison of the Roget and LLM directed knowledge graphs. Top central nodes are ranked by PageRank.

Metric	Roget Graph	LLM Graph
Nodes	2,708	1,066
Edges	2,694	1,093
Communities (Louvain)	29	28
Modularity Q	0.948	0.944
Node Jaccard similarity J	0.028	
Edge Jaccard similarity J	0.015	
LLM-only nodes (source-nonmatched)	1,005 (94.3 %)	
Top 5 Nodes by PageRank		
1	dextrality	brown
2	right	tawny
3	right hand	hazel
4	dexter	chestnut
5	offside	mahogany

The node-set Jaccard similarity of 0.028 indicates near-complete divergence between the two graphs. It means that only 2.8 % of nodes are shared between the two graphs: the LLM-generated and Roget graphs are, for all practical purposes, disjoint knowledge structures. Of the 1,066 nodes in the LLM graph, 1,005 (94.3 %) are source-nonmatched terms with no counterpart in the 1911

text.

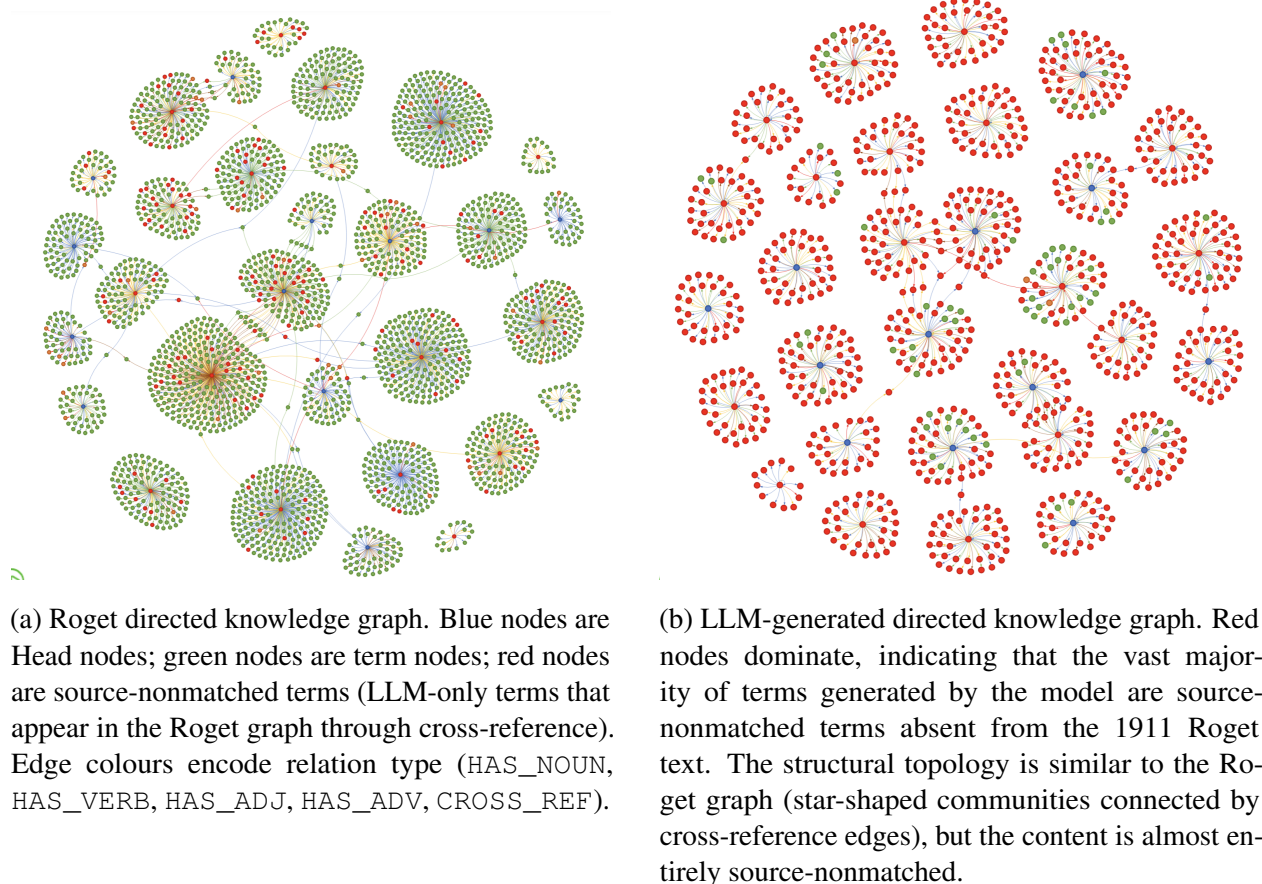


Figure 2: Directed knowledge graphs for the Roget (left) and LLM-generated (right) datasets. Each cluster represents one of the 30 sampled Heads. The visual similarity of the two graphs conceals a near-total divergence in node content: the LLM has reproduced a broadly similar graph form while replacing virtually all of its substance with source-nonmatched terms. This is the defining visual signature of structural hallucination.

The visual comparison of the two graphs (Figure 2) reveals a striking paradox: the LLM-generated graph appears to be *structurally similar* to the Roget graph (star-shaped communities connected by cross-reference edges, similar modularity $Q \approx 0.944$ vs. 0.948) but *content-divergent* (94.3 % source-nonmatched nodes). The model has learned the *form* of a thesaurus while being unable to reproduce its *substance*. This is precisely the mechanism of structural hallucination described in Section 3: the model generates output that is structurally plausible while being source-nonmatched.

Centrality Analysis and Conceptual Re-centering. The centrality comparison reveals the second failure mode: conceptual re-centering.

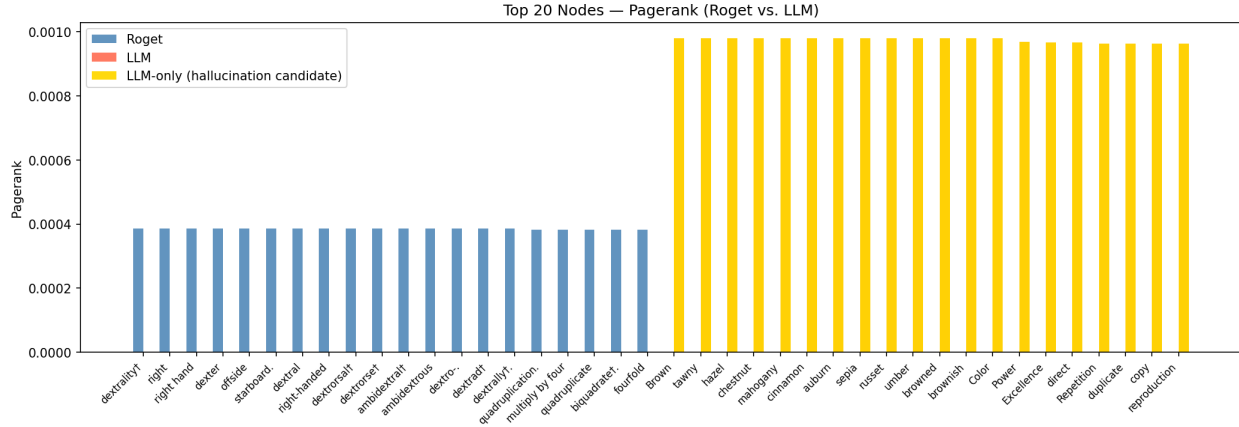


Figure 3: PageRank comparison for the displayed Roget and LLM-only node sets. Blue bars denote Roget nodes, and gold bars denote LLM-only nodes, labelled in the plot as “LLM-only (source-nonmatched).” The legend also includes an orange LLM category, but no orange bars are displayed in this rendering. The colours encode source provenance and matching status; they do not by themselves establish semantic invalidity or causal structural effects.

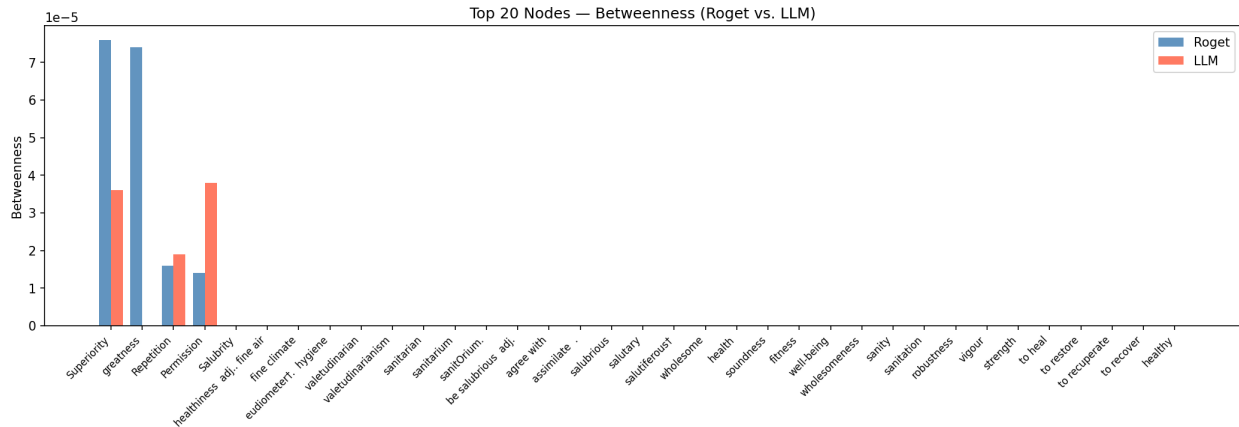


Figure 4: Betweenness centrality for the non-zero values among the displayed Roget and LLM nodes. Blue bars show Roget values and orange bars show LLM values. The re-exported panel retains the categories with at least one non-zero value and omits categories with zero values in both series. Colours identify graph source only; they do not, by themselves, establish source-mismatch, gatekeeper status, or a causal structural effect.

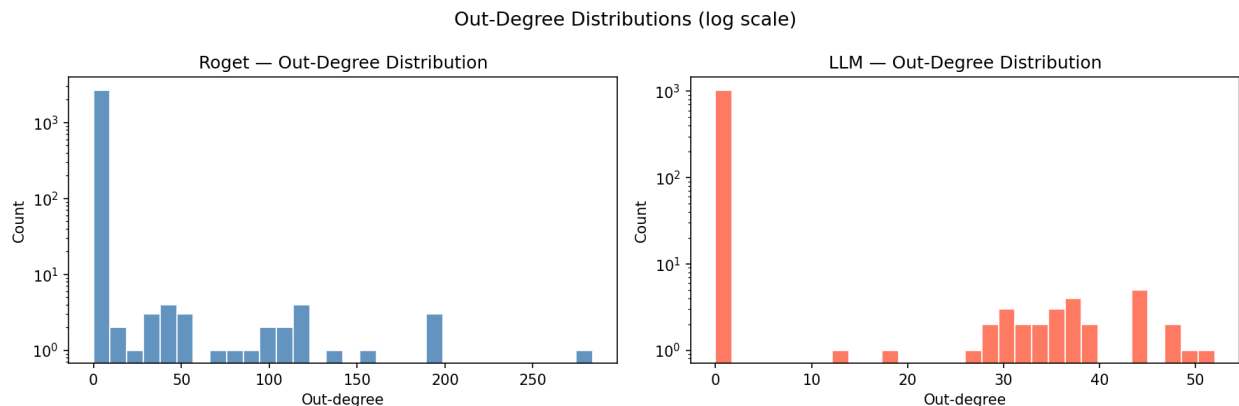


Figure 5: Degree distributions for the Roget (left) and LLM-generated (right) knowledge graphs. Both distributions follow a power-law-like pattern characteristic of hub-and-spoke ontological structures, consistent with scale-free network models (Barabási and Albert, 1999). However, the LLM-generated graph has a lower maximum degree and a different hub composition: the highest-degree nodes in the LLM-generated graph are predominantly source-nonmatched terms rather than the canonical Head nodes that dominate the Roget graph.

The PageRank and betweenness centrality comparisons (Figures 3 and 4) expose the mechanism of conceptual re-centering. In the Roget graph, centrality is concentrated on Head nodes, which are the documented conceptual anchors of the reference structure. In the LLM graph, source-nonmatched term nodes are elevated to high-centrality positions, effectively displacing the canonical conceptual structure. The model alters the centrality hierarchy of the knowledge graph.

Table 3: Top 10 nodes by PageRank centrality in the Roget reference graph and LLM-generated graphs. All LLM nodes shown are absent from the 1911 Roget text.

Roget Top 10 (PageRank)	LLM Top 10 (PageRank)
dextrality	brown
right	tawny
right hand	hazel
dexter	chestnut
offside	mahogany
starboard	cinnamon
dextral	auburn
right-handed	sepia
dextrorsal	russet
dextrorse	umber

Table 3 makes conceptual re-centering explicit. In the reference graph, centrality is concentrated on canonical Head nodes related to the semantic field of “right” and spatial orientation. In contrast, the LLM-generated graph elevates a cluster of colour terms—none of which exist in the original Roget text—to the highest-centrality positions. The upward mobility of exclusively source-nonmatched nodes provides a direct operational demonstration of structural hallucination: source-nonmatched concepts occupy the most structurally influential positions in the generated graph.

8.1.3 Historical Lexical Fidelity and Temporal Bias

The lexical divergence in Benchmark 1 reflects a systematic failure of historical lexical fidelity. The 1911 Roget’s *Thesaurus* encodes a period-specific knowledge organization system, which the LLM replaces with statistically dominant modern synonyms (e.g., replacing terms for right-handedness like *dextrality* with colors like *brown*). This substitution pattern suggests that structural hallucination is a consequence of temporal distributional bias in the training data, where modern linguistic usage is overrepresented. The model interpolates toward present-day lexical regularities, resulting in an anachronistic reconstruction that preserves structural form while overwriting historical content. This “ontological drift” (Tennis, 2002) represents a shift from domain-specific epistemic organization to corpus-wide distributional generalization, where the model optimizes for statistical plausibility rather than fidelity to the historical logic of the knowledge system.

Table 4: Illustrative lexical divergence between the 1911 Roget text and LLM-generated replacements. Examples drawn from the sampled Head on spatial orientation.

Roget 1911 Reference Term	LLM-Generated Term
dextrality	brown
dextrorsal	hazel
dextrorse	russet
dextral	chestnut
right hand	mahogany
starboard	cinnamon
offside	tawny

Table 4 provides qualitative grounding for the quantitative source-mismatch rate. The Roget terms belong to a coherent semantic field of spatial orientation, while the LLM-generated terms cluster around contemporary colour descriptors that are statistically salient in modern corpora but

semantically unrelated to the original Head.

This finding has direct implications for any application that uses LLM-generated knowledge graphs for downstream tasks such as semantic search, concept recommendation, or ontology alignment (Ehrlinger and Wöß, 2016; Paulheim, 2017). The most structurally influential nodes in the LLM-generated graph are precisely the ones that are absent from the reference structure.

This pattern can be further interpreted through Hjørland’s domain-analytic approach, which holds that knowledge organization systems must reflect the historically situated epistemic norms of specific scholarly domains rather than abstract or universal semantic regularities (Hjørland, 2002, 2017). From a domain-analytic perspective, Roget’s 1911 knowledge organization system embodies the conceptual structure, classificatory logic, and lexical conventions of its historical moment. The LLM’s replacement of period-specific terms with statistically dominant contemporary synonyms therefore represents not merely lexical substitution but a failure to respect domain-bound historical semantics. The resulting “ontological drift” reflects a shift from domain-specific epistemic organization to corpus-wide distributional generalisation. In this sense, structural hallucination can be understood as a breakdown in domain-sensitive representation: the model optimises for cross-domain statistical plausibility rather than fidelity to the internal historical logic of a particular knowledge organization system.

8.2 Benchmark 2: Biographical Knowledge — Wikidata Philosophers

The second benchmark evaluates the LLM’s capacity to reconstruct structured biographical knowledge for 1,532 philosophers born between 1800 and 1850, sourced from Wikidata (Vrandečić and Krötzsch, 2014). After matching LLM responses to Wikidata entries, 1,527 matched records were available for evaluation. The seven fields evaluated are: birth date, death date, place of birth, country of citizenship, *influenced_by*, and field of work.

The record-level comparison in Section 5.3 illustrates how one source-to-output field difference is represented before the aggregate results below are calculated across the full matched set.

8.2.1 Results

Table 5: Classical evaluation results for Wikidata philosopher reconstruction ($n = 1,527$ matched records). Hallucination rate is the proportion of LLM-generated values classified as hallucinations (absent from both Wikidata and DBpedia).

Field	Precision	Recall	F1	Accuracy	Hall. Rate
birth	0.258	0.183	0.214	–	0.988
death	0.262	0.185	0.217	–	0.991
place_of_birth	0.144	0.129	0.136	–	0.938
country_of_citizenship	0.585	0.588	0.586	–	0.974
influenced_by	0.092	0.117	0.103	–	0.987
field_of_work	0.185	0.326	0.236	–	0.986

LLM vs Wikidata — Philosopher Evaluation (1527 matched philosophers)

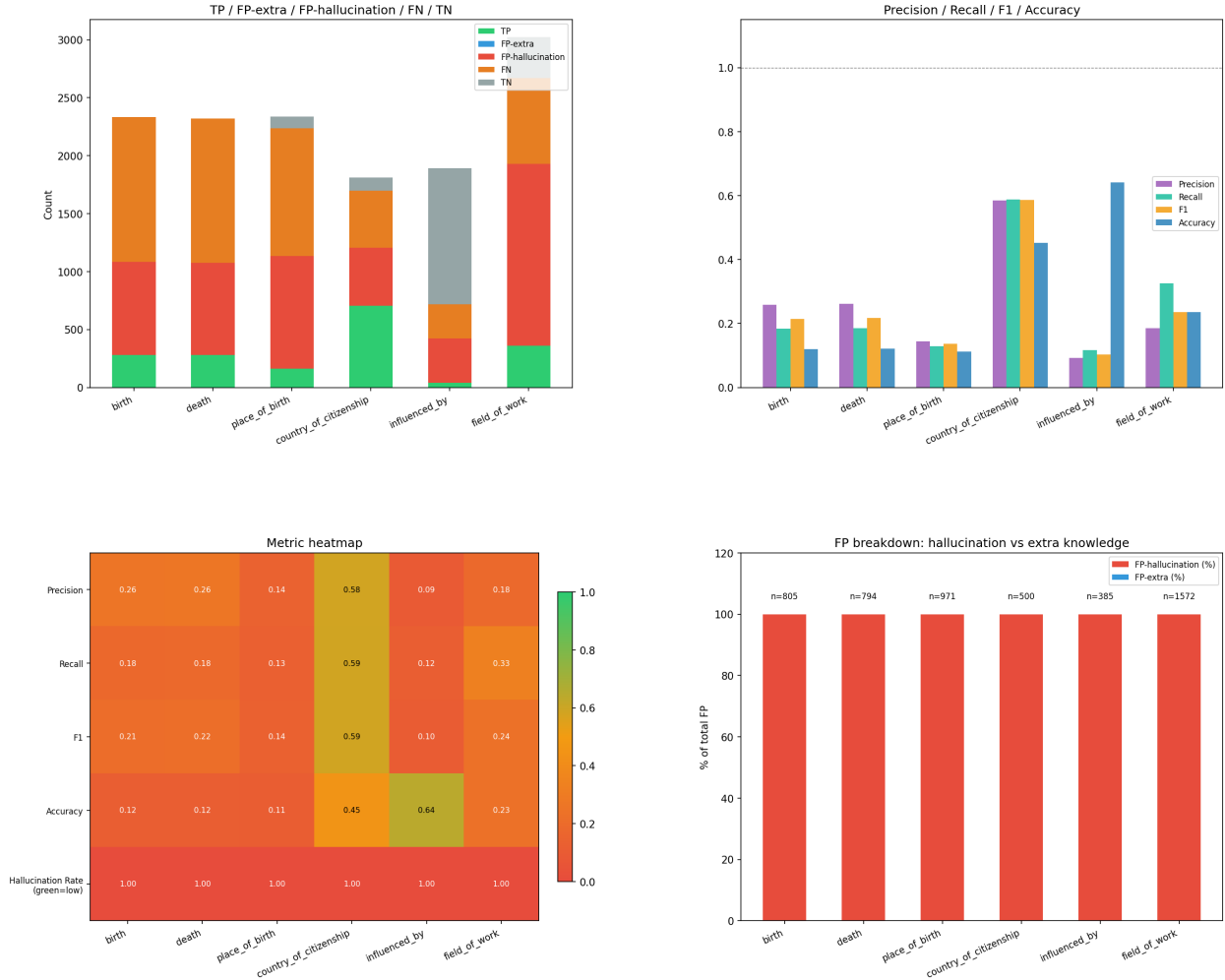


Figure 6: Classical evaluation of LLM reconstruction of Wikidata philosopher data ($n = 1,527$ matched records). The heatmap shows precision, recall, F1, and accuracy per field. Country of citizenship achieves the highest F1 (0.586), reflecting the model’s stronger performance on categorical fields with a small value space. *Influenced_by* achieves the lowest F1 (0.103), reflecting the model’s inability to accurately recall the specific intellectual lineages of 19th-century philosophers. Hallucination rates exceed 93 % for all fields.

The results reveal a consistent pattern: the model performs better on categorical fields with a small, well-defined value space (country of citizenship: F1 = 0.586) and worse on set-valued fields requiring specific factual recall (*influenced_by*: F1 = 0.103; place of birth: F1 = 0.136). Hallucination rates exceed 93 % across all evaluated fields, ranging from 93.8 % (place of birth) to 99.1 % (death date).

The death date result is particularly striking. The model achieves an F1 of only 0.217 for

death dates, despite this being a scalar field with a single correct answer. The hallucination rate of 99.1 % means that virtually every death date provided by the model is absent from both Wikidata and DBpedia. The low F1 and high hallucination rate suggest generative approximation rather than factual retrieval.

The *influenced_by* field highlights the model’s weakness on relational knowledge. With $F1 = 0.103$ and a hallucination rate of 98.7 %, the generated influence networks are largely inconsistent with the historical record. This illustrates structural distortion in relational biographical fields, consistent with the mechanism described in Section 3.

Representative source–output comparisons. Two cached records make the field-level comparison concrete. For Kuno Fischer, the Wikidata reference record gives the birth date 1824-07-23, whereas the cached reconstruction gives 1824-04-22; the values do not satisfy the local date-matching rule. The reference *influenced_by* field is empty, whereas the reconstruction supplies *Hegel* and *Kant*; both are local reference nonmatches. Thus, relative to this reference record, the output substitutes a birth-date attribute value and adds two relation targets.

For Carl Stumpf, the cached birth date matches exactly (1848-04-21), but the reference *influenced_by* value is *Bernard Bolzano* whereas the reconstruction gives *Franz Brentano*. The generated relation value does not satisfy the local comparison against the reference value, so it replaces the recorded relation target in the field–value representation. These examples are source-reference mismatches under the recovered local rule. Because the retained evaluation cache does not preserve item-level DBpedia adjudications, they do not independently establish whether any unmatched generated value is verified extra knowledge, factually false, or a limitation of the specified reference record.

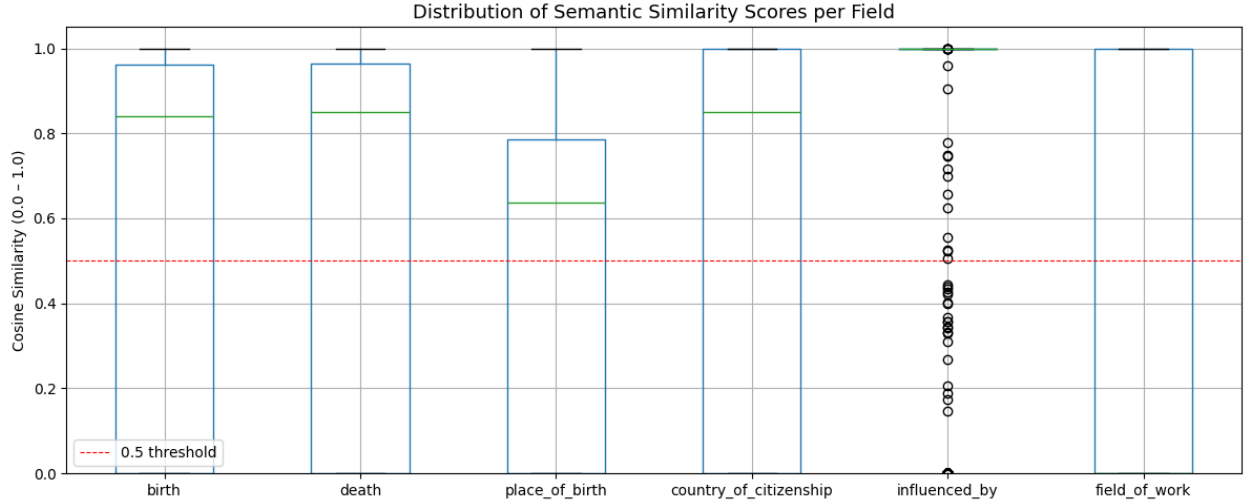


Figure 7: Semantic similarity scores (cosine similarity using all-MiniLM-L6-v2) for the philosopher benchmark. Even for fields where the model achieves moderate F1 scores, the semantic similarity distributions are wide and low, indicating high variance in the quality of LLM output across philosophers. The wide distributions for *influenced_by* and *field_of_work* reflect the model’s inconsistent handling of relational and categorical fields.

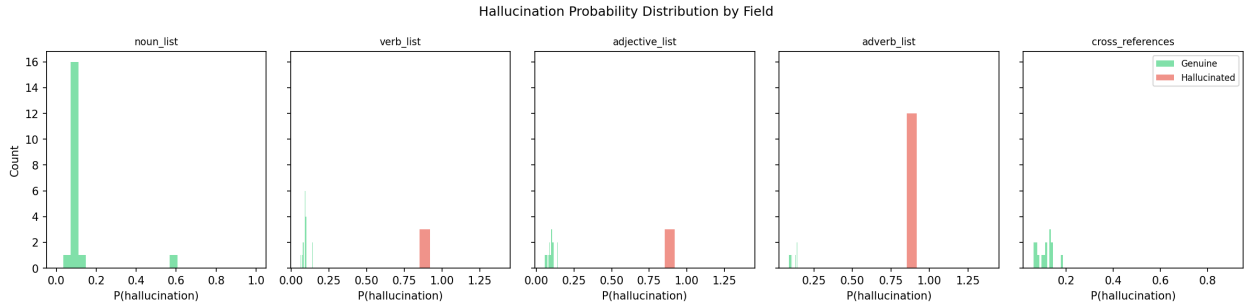


Figure 8: Hallucination probability per field for the philosopher benchmark, as assigned by the logistic regression hallucination classifier. The classifier assigns high hallucination probability to birth and death dates, *influenced_by*, and field of work — precisely the fields that require specific factual recall rather than categorical assignment. The near-zero probability for country of citizenship reflects the model’s relatively reliable performance on this categorical field.

8.3 Benchmark 3: Bibliographic Knowledge — Citations Retrieved from Dimensions.ai

The third benchmark evaluates the LLM’s capacity to reconstruct bibliographic data for 50 COVID-19 publications and their 654 citations, sourced from the Dimensions.ai API (Digital Science, 2026) via the `dimcli` Python library. For each one of these publications, the ten fields evaluated are:

publication authors, title, year, date, DOI, type, concepts, research areas, times cited (i.e., the number of times the publication has been cited by other publications), and the full list of citations (references).

8.3.1 Citation Reconstruction Results

For each publication in this reference bibliographic dataset, we construct a standardized reference string concatenating the following three fields: authors, title, and year. That string is then sent to the LLM with a strict instruction to return a JSON object containing seven specific bibliographic fields (date, DOI, type, concepts, research areas, times cited, and citations). After the model’s response is parsed, cleaned, type-normalized, and converted into structured fields, we obtain an LLM-generated bibliographic record that can be directly aligned with and compared to the reference dataset.

The central quantitative finding of the bibliographic benchmark is the extremely low recall of the citation records. The reference dataset contains 654 unique citations across 50 publications. The LLM returned only 53 citations in total — a recall rate of 8.1 %. More strikingly, the model attempted to provide any citations at all for only 14 of the 50 publications (28 %), leaving the remaining 36 publications (72 %) entirely without citations.

Table 6: Citation recall statistics for the bibliographic benchmark.

Metric	Value
Total reference citations (Dimensions)	654
Total LLM-generated citations	53
Recall rate	8.1 %
Omission rate	91.9 %
Papers with > 0 LLM citations	14 / 50 (28 %)

This pattern indicates a structural limitation rather than a prompting artifact. Within this benchmark, the citation records were not reliably recovered through the tested parametric-generation procedure. As argued in Section 3, this is reported as bibliographic structural divergence: the model does not reliably reproduce the reference citation records.

Representative citation-network comparisons. Two cached records make the aggregate result concrete. For *The contribution of the private healthcare sector during the COVID-19 pandemic: the experience of the Lombardy Region in Northern Italy.*, the reference snapshot contains 19 citing records, including *The energy burden of care: Insights from a regional survey in Italian hospitals.* (10.1177/09514848261416445), whereas the cached LLM reconstruction returns

no citations. Relative to the experimental reference snapshot, this omits all 19 observed citation edges for that record.

For *Small fibre neuropathy frequently underlies the painful long-COVID syndrome*, the reference snapshot contains 18 citations, including *Non-invasive electrodiagnostic testing for small fibre neuropathy in long COVID-19* (10.1136/bmjno-2025-001399). The cached reconstruction returns five citations, including *Neuropathic manifestations in post-COVID-19 patients: A systematic review* (10.1016/j.jns.2024.120123). The retained evaluation records a mean maximum title similarity of 0.757 across the five generated citations, a year-match rate of 1.0, and a DOI-validity rate of 0.0. This is a source–output citation-set and identifier mismatch under the cached comparison; it does not establish that every generated title is fictitious or that each unresolved DOI is an unresolved identifier.

8.3.2 Field-Level Results

Table 7: Classical evaluation results for the Dimensions bibliographic benchmark ($n = 50$ publications).

Field	Precision	Recall	F1	Accuracy	Hall. Rate
date	0.39	0.44	0.39	0.18	≈ 0
doi	0.01	0.01	0.01	0.01	≈ 1.0
type	0.70	0.70	0.70	0.54	≈ 0
time_cited	0.06	0.04	0.06	0.06	high
main_concepts	0.22	0.23	0.23	0.11	0.20
research_areas	0.01	0.01	0.01	0.01	0.20

The field-level results differ across the reported benchmark variables. In the recovered 50-record source cache, publication type takes exactly three observed normalised values: `article` (30 records), `book` (10), and `chapter` (10). The corresponding cached LLM outputs also use these three labels—`article` (45), `book` (3), and `chapter` (2)—so the publication-type result ($F1 = 0.70$) is evaluated within this sample-specific three-label set. This observed set is not claimed to be a complete Dimensions taxonomy. The retained materials do not establish a finite controlled range for date, DOI, times cited, main concepts, or research areas; their lower scores are therefore reported without attributing them to an unverified vocabulary size, hierarchy, or inherent inferability from title and year.

The DOI failure is particularly instructive. A DOI is a unique, persistent identifier that cannot be guessed or inferred; it must be looked up in a registry (Paskin, 2010). The model’s near-zero F1 score for DOIs confirms that it is not retrieving identifiers from memory but generating

LLM vs Dimensions — Bibliographic Evaluation (50 matched publications)

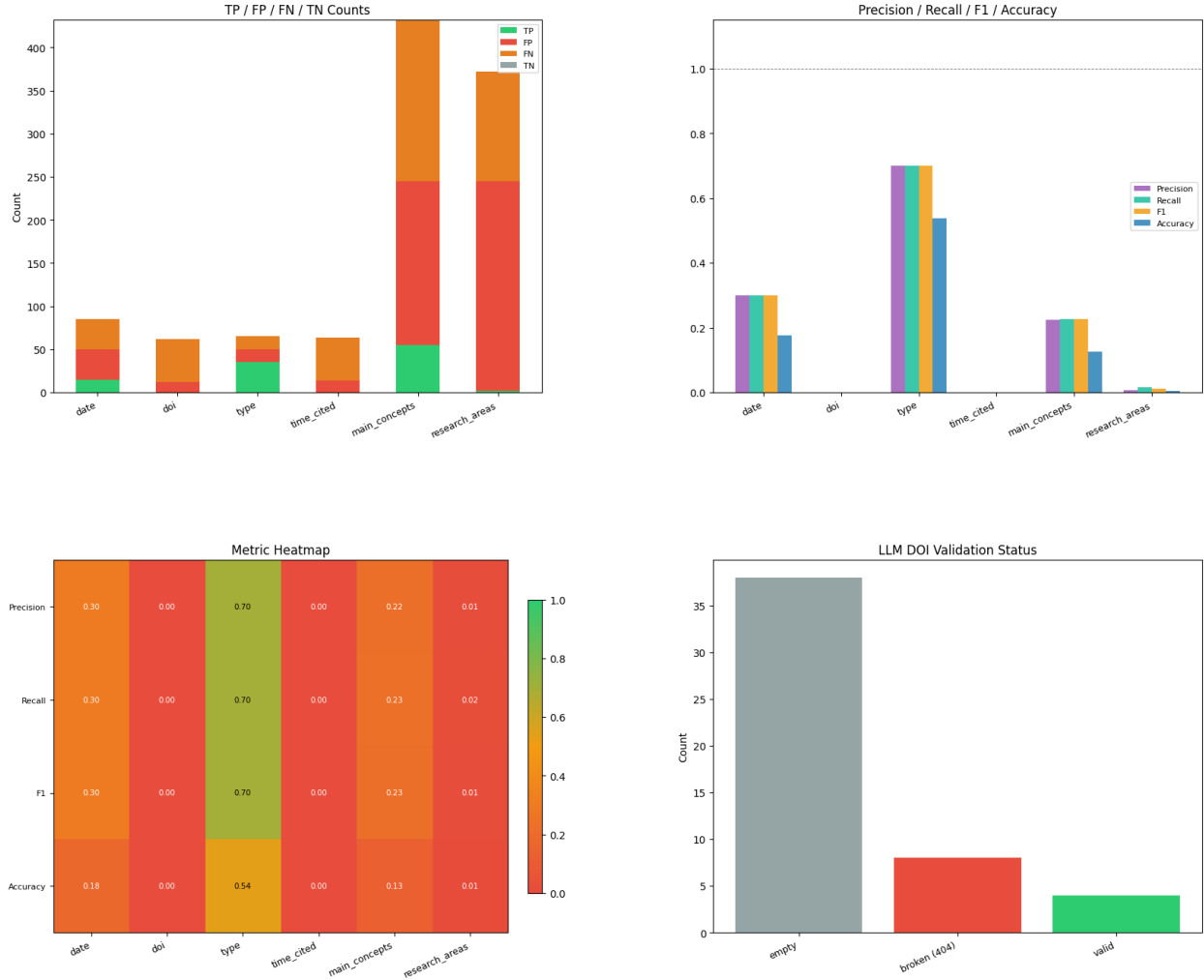


Figure 9: Classical evaluation of LLM reconstruction of Dimensions bibliographic data ($n = 50$ publications). The four panels show (top left) TP/FP/FN/TN counts per field, (top right) precision/recall/F1/accuracy per field, (bottom left) a metric heatmap, and (bottom right) DOI validation status. The DOI validation panel is particularly revealing: the majority of LLM-generated DOIs are either empty or broken (invalid format), with only a small fraction being valid — and even valid DOIs are rarely correct.

plausible-looking strings that happen to be wrong. The hallucination rate for DOIs approaches 100 %: virtually every DOI provided by the model is an unresolved identifier.

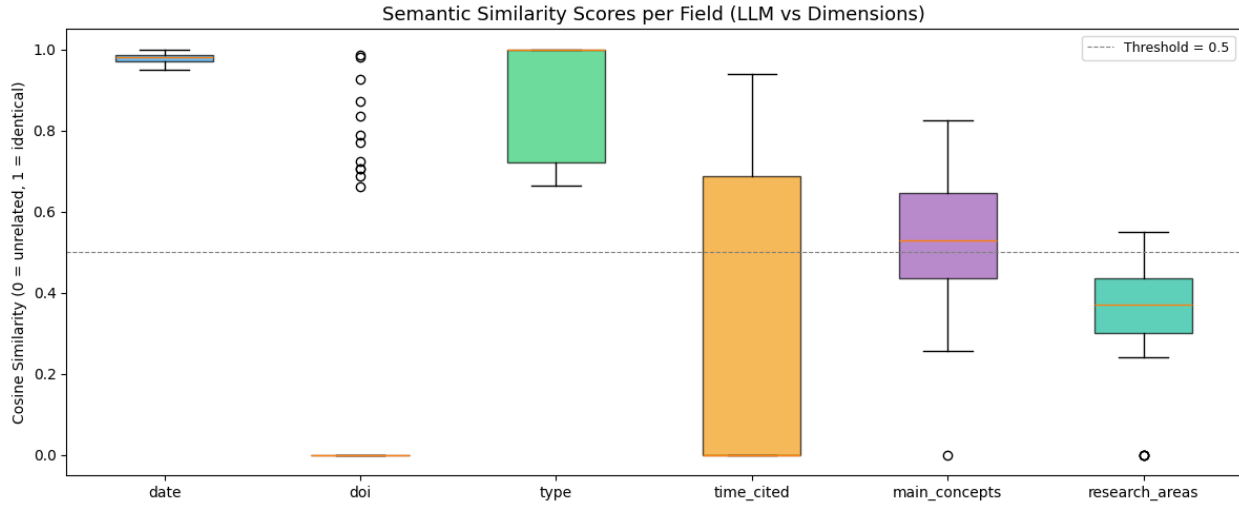


Figure 10: Distribution of cosine-similarity scores by field for LLM-generated and reference Dimensions values, computed with `all-MiniLM-L6-v2`. Scores range from 0 (unrelated) to 1 (identical). The boxplots show the distributions for `date`, `DOI`, `type`, `time_cited`, `main_concepts`, and `research_areas`; horizontal median lines, boxes, whiskers, and circular outliers follow the standard boxplot convention. The dashed horizontal line marks the similarity threshold of 0.5.

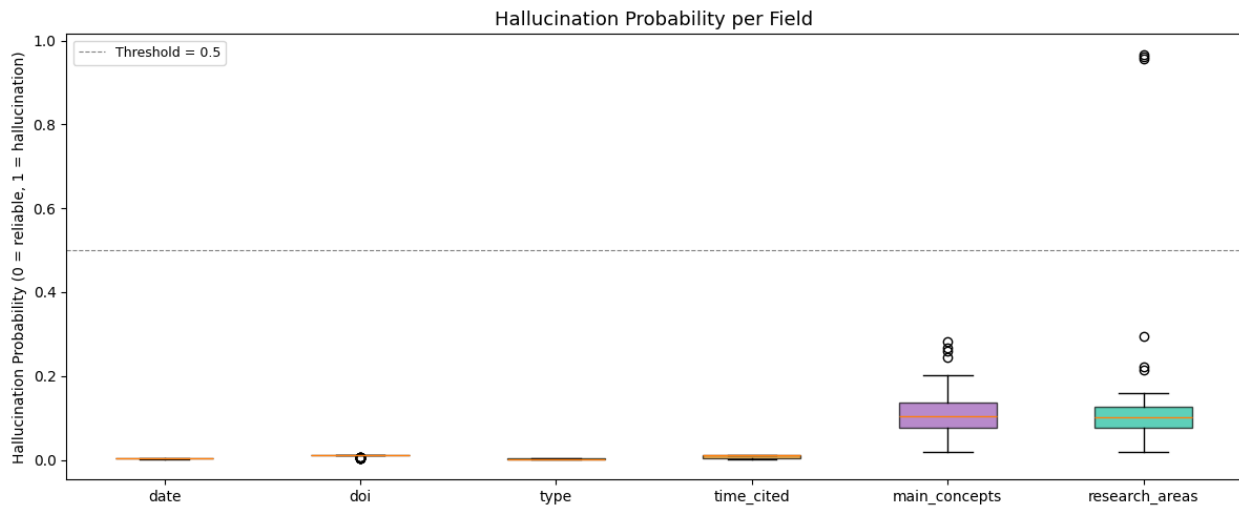


Figure 11: Hallucination probability per field for the bibliographic benchmark, as assigned by the logistic regression hallucination classifier. The classifier assigns the highest hallucination probabilities to `times_cited`, `DOI`, `research_areas`, and `main_concepts`. These are precisely the fields that require access to external databases or real-time citation counts — information that is structurally inaccessible to a language model operating from training data alone.

8.4 Cross-Domain Structural Pattern

Across the three benchmarks, a comparable pattern emerges under the stated evaluation procedures. In the lexical domain, the language model reproduces ontological form while replacing historical vocabulary with semantically related contemporary terms. In the biographical domain, performance is relatively stronger for categorical fields with constrained value spaces and substantially weaker for relational fields requiring historically grounded network knowledge. In the bibliographic domain, identifier-based and citation-network fields exhibit extremely low recall, indicating non-retrievability of database-structured relations.

Taken together, these results suggest that parametric language models generate semantically plausible approximations without reliably reconstructing externally verifiable relational structures. Classical field-level metrics capture local mismatch, but graph-based diagnostics reveal structural divergence that is not visible through token-level evaluation alone. This cross-domain consistency supports the structural hallucination hypothesis advanced in Section 3.

9 Discussion

Table 8: Cross-domain structural divergence summary across lexical, biographical, and bibliographic benchmarks.

Domain	Source Mismatch / Operational Hallucination Label	Jaccard Similarity	Citation O
Lexical (Roget)	94.3% source-nonmatched nodes	0.028	–
Biographical (Wikidata)	>93% hallucinated fields	–	–
Bibliographic (Dimensions)	High (DOI $\approx$ 100%)	–	91.9

Table 8 summarizes the divergences observed under the study’s matching and evaluation procedures in three selected benchmark settings. The Roget case should be interpreted as a historical lexical stress test: its differences may reflect both non-recovery of the 1911 source structure and historical change in terminology. The Wikidata and Dimensions analyses provide separate contemporary tests involving biographical and bibliographic data. Across these cases, `gpt-4.1-mini` outputs show low alignment with the respective benchmark-defined relational structures, expressed as node or attribute mismatches and citation omissions. This convergence supports the usefulness of structural hallucination as a descriptive framework for the mismatches observed here; it does not establish a universal property of all knowledge organization systems, model architectures, or task settings.

Across the three benchmarks, the evaluated language model produced outputs that were often surface-plausible but did not reliably align with the benchmark-defined relational structures. In the historical Roget stress test, this mismatch must be interpreted in light of both non-recovery of the 1911 source structure and historical lexical change. Importantly, the biographical (Wikidata) and bibliographic (Dimensions) benchmarks provide independent contemporary evaluations, indicating that structural divergence is not confined to the historical lexical case. Taken together, these results demonstrate the value of examining structural divergence alongside local accuracy within this evaluation design. They support structural hallucination as a useful description of the observed failures, but do not establish that such divergence is universal across parametric models, knowledge organization systems, or task settings, or that it would persist under retrieval-augmented generation.

The empirical findings have three methodological implications.

First, LLM outputs should not be treated as substitutes for structured database retrieval when precise bibliographic or factual reconstruction is required. The observed source-mismatch and omission rates indicate that direct verification against authoritative sources remains necessary.

Second, semantic similarity does not guarantee structural fidelity. Moderate semantic alignment in the Roget benchmark coexists with near-total node-set divergence. This confirms the analytical distinction between local plausibility and global relational alignment introduced earlier.

Third, network-analytic comparison provides diagnostic information not captured by classical precision–recall metrics. Measures such as node-set Jaccard similarity, centrality rank correlation, and modularity comparison expose structural divergence that would remain invisible under sentence-level evaluation alone.

9.1 Limitations and Scope Conditions

Several limitations bound the interpretation of these findings:

- **Reference bias and domain dependence.** The reference graphs function as documented comparison baselines rather than as absolute truth. They embody source-specific scope, provenance, curatorial decisions, versioning, and possible omissions or biases. In domains without a canonical graph, researchers must construct a transparent expert reference or use consensus-based comparison. The present results may differ for other domains, model architectures, retrieval-augmented systems, or task formulations.
- **Historical bias.** The Roget analysis is a historical lexical stress test based on the 1911 edition. Its observed mismatch reflects both failure to recover the source structure and historical change

in terminology. It does not provide independent evidence about contemporary knowledge organization systems.

- **Matching sensitivity.** The strict operational matching used here may penalize semantic variation. The fuzzy matching threshold may undercount partial lexical correspondence, and the DBpedia oracle is not exhaustive. All results are conditional on the stated matching thresholds; alternative matching rules or semantic embedding thresholds would yield different absolute mismatch rates.
- **Contemporary KOS coverage.** Wikidata and Dimensions provide contemporary knowledge-graph and bibliographic-graph cases, but they do not substitute for a contemporary KOS-specific replication. A benchmark using a current controlled vocabulary, thesaurus, or ontology is needed to isolate structural reconstruction from historical lexical change.

These limitations restrict the present claims to the evaluated model, benchmarks, and evaluation design while identifying priorities for replication and sensitivity analysis.

10 Conclusion

This paper advanced a conceptual account of structural hallucination—distortion at the level of relational organization—and a set of empirical benchmarks to detect it. We argued that sentence-level verification is inadequate for domains where knowledge is organized relationally and introduced a reproducible validation architecture grounded in knowledge graph extraction, network-analytic diagnostics, and citation integrity verification.

Within the three benchmark settings examined here, `gpt-4.1-mini` outputs showed substantial divergence from the documented comparison structures: limited node-set overlap in the historical Roget stress test, high biographical field mismatch in Wikidata, and citation omission in Dimensions. The Roget result must be interpreted in light of both non-recovery of the 1911 source structure and historical change in terminology. These findings show that semantic plausibility need not guarantee structural fidelity under the study’s extraction and matching procedures, and that graph-level evaluation can complement local accuracy metrics when a documented reference structure is available.

Beyond research evaluation, the framework has direct implications for digital library governance. As libraries experiment with AI-assisted metadata generation, structural hallucination risks propagating into catalogues and discovery layers. The hallucination stress test can be operationalised as a pre-ingestion quality-control layer: curators could extract a knowledge graph from

generated records, compare it against authoritative reference graphs, and compute source-mismatch and omission rates. Records exceeding structural deviation thresholds would trigger human review, functioning analogously to existing authority control procedures (Borgman, 1999; Tennis, 2012). Such integration would transform LLM oversight from reactive correction to proactive infrastructural design. The framework and benchmarks presented here provide a methodological basis for evaluating structural alignment in LLM-generated scholarly representations. Future research may extend the approach to additional domains, incorporate retrieval-augmented architectures, and explore the interaction between parametric generation and external knowledge graphs. The broader challenge is not whether LLMs can produce fluent text, but how their outputs can be assessed for structural compatibility with established knowledge systems.

Generative AI Use Statement

Uses and Roles of Generative AI: The manuscript evaluates structured outputs produced by gpt-4.1-mini in the reported benchmark experiments. A generative-AI assistant was also used during preparation of the revision for drafting, formatting, and code-documentation assistance. The author reviewed and approved all AI-assisted material and remain fully responsible for the manuscript's content.

Data Availability Statement

The code-only reproducibility package for this study is maintained at https://github.com/mboudour/Network_Based_LLMs_Hallucination_Eval (private repository; editorial and reviewer access is available from the authors on request). It contains the analysis code, fixed benchmark prompts, model-configuration information, normalization and matching routines, figure-generation scripts, and documentation needed to inspect the study workflow.

The repository intentionally omits source and processed benchmark data, cached provider records, raw LLM outputs, query logs, item-level matching records, and figure files. The principal restriction concerns the Dimensions benchmark: Dimensions bibliographic and citation records are governed by provider access and redistribution terms. The cached source records, query logs, and record-level matching materials are therefore not publicly deposited where they reproduce, or could permit reconstruction of, restricted provider content. The corresponding raw LLM outputs are retained with that controlled record-level evaluation archive rather than released as a separate public dataset. These restrictions concern source licensing and provenance; they do not prevent

methodological inspection of the code, prompts, configurations, normalization rules, and figure-generation workflow.

The Roget source text is available from Project Gutenberg (<https://www.gutenberg.org/ebooks/10681>), and Wikidata reference records can be re-queried through the Wikidata Query Service (<https://query.wikidata.org>). The rendered figures are included in the manuscript and can be regenerated from the deposited code by users with the relevant source access. Where permitted by the applicable source terms, the controlled archive may be made available to the editor and reviewers on request.

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